INCH-POUND

MIL-A-82845(OS)
5 February 1990
SUPERSEDING
WS 26952
14 December 1987

MILITARY SPECIFICATION

ADHESIVE, PASTE, ONE-PART, NON-ASBESTOS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a non-asbestos one-part paste adhesive, referred to herein as the adhesive. The adhesive contains no asbestos and consists of a modified epoxy resin.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-T-81533 1,1,1-Trichloroethane (Methyl Chloroform) Inhibited, Vapor Degreasing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, Attn: 1053, 5801 Tabor Avenue, Philadelphia, PA 19120-5094.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

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NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1824 Apparent Viscosity of Plastisols and Organosols at Low Shear Rates by Brookfield Viscometer (DoD adopted)

ASTM D 1876 Peel Resistance of Adhesive (T-Peel Test) (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of the adhesive shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. The adhesive shall be a one-part homogeneous paste exhibiting a light tan color.

3.2.1 Asbestos. The adhesive material shall contain no asbestos (see 3.8 and 4.1.2).

3.3 Physical properties. Physical properties of the adhesive shall conform to table I.

3.4 Infrared identification. The adhesive infrared identification shall conform qualitatively to figure 1.

3.5 Storage temperature. The maximum temperature to which the adhesive shall be exposed during storage shall be 0°F.

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TABLE I. Physical properties.

Property	Limits	
	Minimum	Maximum
Viscosity, poise		
76 ± 1°F	3000	20000
Overlap shear, psi		
75 ± 5°F	4500	---
180 ± 5°F	3200	---
250 ± 5°F	500	---
T-peel, lb/inch		
75 ± 5°F	25	---

3.6 **Age of adhesive.** The adhesive shall be inspected as specified in section 4 not more than 90 days prior to shipment.

3.7 **Shelf life.** The adhesive shall have a shelf life of 9 months after date of inspection (see 3.6) when stored at 0°F or lower in original closed containers. The maximum cumulative time out of controlled storage shall be 36 hours.

3.8 **Safety.**

3.8.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.8.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.9 **Workmanship.** The adhesive component materials shall be free from agglomeration, foreign matter, separation, dilution, and contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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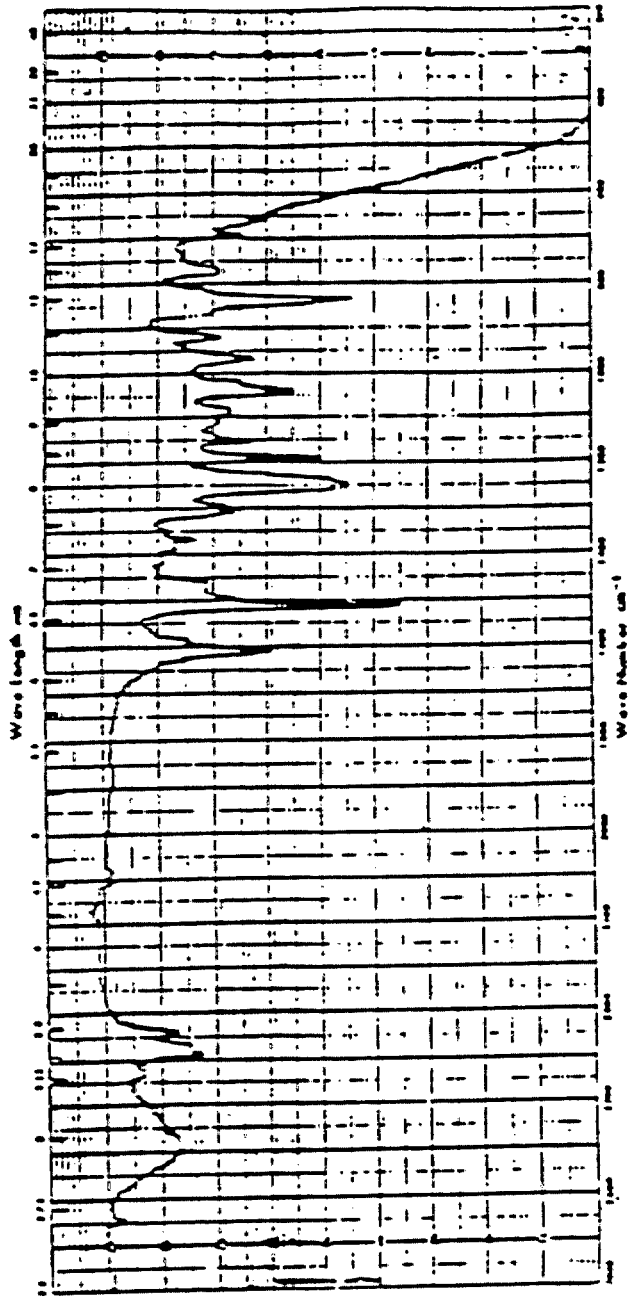


FIGURE 1. Adhesive infrared identification.

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4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspection (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. Quality conformance inspection.

Examination and Tests	Requirement	Inspection Method
Visual examination	3.2 and 3.9	4.5.1
Viscosity	3.3	4.5.2
Overlap shear	3.3	4.5.3
T-peel	3.3	4.5.4
Infrared identification	3.4	4.5.5
Packaging	Section 5	4.5.6

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4.4.1 Lot. A lot of adhesive shall consist of all material which is compounded and mixed in a single container in one mixing period without change in process or materials and submitted for inspection at one time. Lot conformance to requirements shall be based on average test results.

4.4.2 Sampling. Sufficient material to conduct three tests for all physical properties specified in 3.3 and one infrared identification shall be removed from each lot of adhesive. Sampling for the visual examination (see 4.5.1) shall consist of the samples selected for the specified tests and infrared identification.

4.4.3 Classification of characteristics. The characteristics verified by the tests and examination herein are classified as Major or Minor in accordance with WR-43. Tests that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the test. The visual examination herein is not annotated with a classification code or serial number and is classified Minor.

4.5 Methods of inspection. The following methods of inspection shall be used. Unless otherwise specified in the applicable inspection method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 Visual examination. Visual examination shall be made to verify conformance to the requirements of 3.2 and 3.9. Contractor certification verifying conformance to 3.2.1, 3.5, 3.6 and 3.8 shall be required (see 6.2).

4.5.2 Viscosity. The viscosity of the adhesive shall be determined in accordance with ASTM D 1824 and the following:

a. Use a Brookfield Model HBF or HBT with a number 7 spindle at 10 rpm.

b. Record individual and average values to the nearest 100 poise.

4.5.3 Overlap shear (M101). Overlap shear shall be determined as specified in 4.5.3.1 through 4.5.3.7.

4.5.3.1 Materials required.

a. 2024 T3 clad 0.063-inch by 4-inch by 7-inch aluminum panels having one edge milled.

b. Owens-Corning 2/2 glass thread oil-starch size.

c. Adhesive (see 4.4.2).

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4.5.3.2 **Preparation of aluminum surface.**

a. Scribe both ends of the burr-free face of milled edge 1/2 inch from the edge.

b. Vapor degrease with perchloroethylene.

c. Alkaline degrease in an agitated solution of 9 to 11 ounces per gallon of Oakite 164 for 10 minutes minimum at 190 ± 10°F.

d. Immediately (before drying occurs) after removal from this bath, dip in a tank of tap water.

e. Flush thoroughly with a spray of clear tap water.

f. Place panels in the following etching solution for 10 minutes minimum at 155 ± 5°F.

FPL Etch Solution

By Volume	By Weight
450 ml water	30 parts water
84 ml sulfuric acid	10 parts sulfuric acid
30 g sodium dichromate	2 parts sodium dichromate

g. Rinse in tap water.

h. Air dry 15 minutes minimum.

i. Force dry 10 minutes minimum at 150 ± 5°F.

j. Use panels within 4 hours.

NOTE: As an alternative procedure, panels may be wiped with MIL-T-81533 methyl chloroform for 2 minutes using clean cheesecloth. Steps b, c, and d may then be eliminated.

4.5.3.3 **Preparation of glass thread spacers.**

a. Dip in acetone.

b. Air dry 1 minute minimum.

c. Cut in 1- to 1 1/2-inch lengths (± 1/4 inch) (three lengths required per bonded panel).

4.5.3.4 **Application of test material.**

a. One-part adhesive: use as received (see 4.4.2).

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b. Apply a 5-to 10-mil coating of the adhesive on each faying surface.

c. Place glass thread spacers on one surface only--one 3/4 inch from edge, and the other in the middle.

4.5.3.5 Bond lay-up. Place first panel in contact with the second and tape in place with cellophane tape, making sure the overlap is 1/2 inch.

4.5.3.6 Cure. The cure equipment shall consist of a hydraulically operated, electrically heated platen press or autoclave.

a. Cure time: 60 to 90 minutes

b. Cure temperature: 250 ± 10°F

c. Cure pressure: 5 ± 2 psi

d. Use a strip of cardboard over and under the bonded area. Place aluminum foil between cardboard and bond area to facilitate cardboard removal from finished bond.

4.5.3.7 Test.

a. Saw panel into 1-inch strips.

b. Randomize specimens so tests at each temperature include specimens from more than one panel.

c. Ambient temperature Shear Test.

(1) Install test specimen in grips of testing machine and apply load at a rate of 1200 to 1400 psi per minute or a crosshead speed at 0.05 inch per minute (distance from end of overlap to end of jaws should be 2.500 ± 0.200 inches). Record load at failure.

(2) Calculate shear strength as follows:

$$\text{Shear strength, psi} = \frac{Lf}{B}$$

Where:

Lf = failure load, lb
B = total bond surface area, inch squared

(3) Record average and individual values to the nearest 10 psi.

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d. Elevated Temperature Shear Test. Elevated temperature shear tests shall be conducted the same as room temperature shear test with the added requirement that a suitable over or furnace shall be provided to maintain the specimens at the specified test temperature ± 5°F. Specimen temperature shall be measured by means of a thermocouple attached to the overlap area. The specimens shall be conditioned in the test environment for not less than 15 nor more than 30 minutes prior to test.

4.5.4 **T-peel (M102).** T-peel shall be determined as specified in 4.5.4.1 through 4.5.4.6.

4.5.4.1 **Materials required.**

a. Adhesive (see 4.4.2).

b. Aluminum 8-inch by 8-inch by 0.032-inch 2024 T3 clad (Reference ASTM D 1876, figure 1).

c. Glass threads; Owens-Corning 2/2 oil starch size.

4.5.4.2 **Preparation of aluminum surface.**

a. Vapor degrease with perchloroethylene.

b. Alkaline degrease in an agitated solution of 9 to 11 ounces per gallon of Oakite 164 for 10 minutes minimum at 190 ± 10 °F.

c. Immediately (before drying occurs) after removal from this bath, dip in a tank of tap water.

d. Flush thoroughly with a spray of clear tap water.

e. Place panels in the following etching solution for 10 minutes minimum.

FPL Etch Solution

By Volume	By Weight
450 ml water	30 parts water
84 ml sulfuric acid	10 parts sulfuric acid
30 g sodium dichromate	2 parts sodium dichromate

f. Rinse in tap water.

g. Air dry 15 minutes minimum.

h. Force dry 10 minutes minimum at 150 ± 5°F.

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i. Panels should be used within 4 hours.

NOTE: As an alternative procedure, panels may be wiped with MIL-T-81533 methyl chloroform for 2 minutes using clean cheesecloth. Steps a, b, c, and d may then be eliminated.

4.5.4.3 Preparation of glass thread spacers.

a. Dip in acetone.

b. Air dry 1 minute minimum.

c. Cut in 8- to 10-inch lengths (± 2 inches) (three lengths required per bonded panel).

4.5.4.4 Application of test material.

a. One-part adhesive: use as received.

b. Apply a 5- to 10-mil coating of the adhesive on each faying surface of the 8-inch by 8-inch panel. Apply the adhesive leaving a 1/2-inch border free of adhesive on three sides of the panel and a 2-inch border on the fourth side. This is to allow for flow-out. Place one glass thread in the center of one panel, perpendicular to the 2-inch border and running the length of the panel. Place the remaining threads parallel to the center thread but 1/2 inch from each edge of the panel. Do this on one faying surface only. Place the panels together and hold in place with adhesive tape.

4.5.4.5 Cure. The cure equipment shall consist of a hydraulically operated, electrically heated platen press.

a. Time: 60 to 90 minutes

b. Temperature: 250 ± 10°F

c. Pressure: 5 ± 2 psi

d. Use a strip of cardboard over and under the bonded area. Place aluminum foil between cardboard and bond area to facilitate cardboard removal from finished bond.

4.5.4.6 Test.

a. Saw panel into 1-inch strips parallel to the direction of the threads and discard the edge specimens.

b. The tensile load shall be applied to the test fixture at a constant rate of 2 to 3 inches per minute.

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c. Record average load required to peel specimen.

d. Calculate and record individual and average peel strength to the nearest pound per inch.

4.5.5 **Infrared identification.** Infrared identification shall be determined in accordance with the following:

a. Apply a drop of the adhesive to the sodium chloride window.

b. Conduct a smear scan between 2.5 and 25.0 micrometers.

c. Compare the spectrum qualitatively with figure 1 and record as conforming or nonconforming to the requirements of 3.4.

4.6 **Inspection of packaging.** The adhesive containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The adhesive shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.8. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture.

d. Shelf life expiration date (see 3.7).

e. Storage conditions. Mark each container with the following:

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"TEMPERATURE SENSITIVE MATERIAL.
DO NOT EXPOSE TO TEMPERATURE HIGHER THAN 0°F."

f. Toxicity warning (if applicable).

5.3 **Shipping temperature.** The adhesive shall not be exposed to temperatures higher than 0°F during shipment.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The adhesive is intended for use in the structural bonding of components of the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 2.1.2 and 2.2).

c. Whether first article inspection is required (see 3.1 and 6.3).

d. Material Safety Data Sheets Required (see 3.8.2 and 6.4).

e. DOD FAR 52.246-15 (see 4.5.1).

f. Packaging if other than as specified in 5.1.

g. Special marking, if other than as specified in 5.2.2.

6.3 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending

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contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Material Safety Data Sheets. MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 8040-N156)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82845	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
ADHESIVE, PASTE, ONE-PART, NON-ASBESTOS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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